

PRAN

1. - PROVJERA DUPLIKATA U POLJU A[]

CRCW

DUPLIKATI = 0

FOR (i IN (1, m-1)) {

PAR (j IN (0, m-1)):

IF (A[j] == A[j+i])

DUPLIKATI = 1

IF DUPLIKATI == 1

BREAK;

}

IF DUPLIKATI <

PRINT "POSTOJE DUPLIKATI"

}

-> SLOŽENOST $O(m)$

REW

PAR (i IN (0, m-1))

B[i] = A[i]

DUPLIKATI[i] = 0

FOR (i IN (1, m-1)) {

PAR (j IN (0, m-1))

IF (A[j] == B[j+i])

DUPLIKATI[j] = 1

} $O(1)$ } $O(m)$

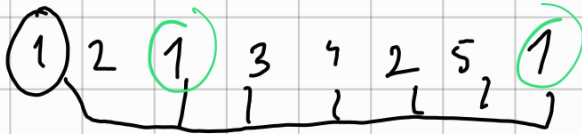
-> SLOŽENOST $O(m)$

REZ = MAX-REDUCE (DUPLIKATI)

} $O(\log m)$

2. - ODREDI BROJ RAZLIČITIH U POLJU P[]

CRCW



REZ = 0

FOR (i in [0, m-1])

UNIQUE = 1

PAR (j in [i+1, m-1])

IF (P[i] == P[i+j])

UNIQUE = 0

REZ += UNIQUE

-> O(m)

PRINT("JEDINSTVENI = " REZ)

EREW

PAR (j in [0, m]) :

B[i] = P[i]

U = [1]

FOR (i in [0, m-1])

PAR (j in [i+1, m-1])

IF (P[i] == B[i+j])

UNIQUE[i+j] = 0

- O(m)

REZ = + REDUCE(UNIQUE)

3. EREW PRAM, PROWIEŻA UZŁAZNIOG SCHEMATY

EREW

PAR(i in $[0, n-1]$)

$B[i] = A[i]$

PAR(i in $[0, n-1]$)

IF ($A[i] > B[i+1]$)

$C[i] = 1$

ELSE

$C[i] = 0$

$REZ = +_REDUCE(C)$

IF ($REZ > 0$)

PRINT("MIE SORTOWAN")

4. - ODRĘBITI KOSI JE NIŻ PRAWI PO ABECEDZI
 $\vee \leq O(m)$ SŁOŻENOSTI

PAR(i in $[0, m-1]$):

IF ($A[i] = B[i]$)

$C[i] = 0$

ELSE IF ($A[i] > B[i]$)

$C[i] = 1$

ELSE

$C[i] = -1$

0 0 $\rightarrow$ 0

0 1 $\rightarrow$ 1

0 -1 $\rightarrow$ -1

-1 * $\rightarrow$ -1

1 * $\rightarrow$ 1

$REZ = OP_REDUCE(C)$

IF ($REZ == 1$) $\rightarrow$ A

IF ($REZ == -1$) $\rightarrow$ B

ELSE $\rightarrow$ IS71

OP(a, b) {

IF ($a == 0$)

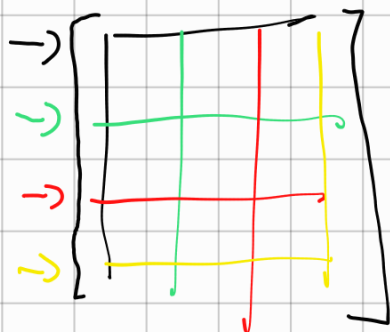
RETURN b

ELSE

RETURN a }

5.

Матрица $\Pi[n][m] \rightarrow$ определить, является ли пермутационной
(в каждом строке и столбце точно одна единица)
Сложность $\leq n^2$



Функция

PERMUTACIJSKA = 1

FOR (i in 1 to n) {

FOR (j in 1 to m) {

R[j] = $\Pi[i][j]$

S[j] = $\Pi[i][j]$

REZ_R = +_REDUCE(R)

REZ_S = +_REDUCE(S)

IF REZ_R != 1 OR REZ_S != 1

PERMUTACIJSKA = 0

BREAK

IF (PERMUTACIJSKA == 1) $\rightarrow$ DA

ELSE $\rightarrow$ NE

$\rightarrow$ Сложность $O(n \cdot m)$

6.

Список vec называется вектором пермутации
(пермутация $1, \dots, n$)

Список

SVI = 1

FOR (i in 1 to n) {

$TR = 0$

$PAR(i \text{ in } [0, m-1])$

$IF (A[i] == i)$

$TR = 1$

$IF (TR == 0)$

$SUM = 0$

$BREAK;$

$IF (SUM) \rightarrow DA$

$ELSE \rightarrow NE$

7. $\rightarrow$ ПРОВЕРКА ПЕРИОДИЧЕСКОГО ПЕРИОДА

ЭЛЕВ

$PAR(i \text{ in } [0, m-1])$

$IF (A[i] == 0)$

$POW_0 = 1$

$PAR(i \text{ in } [0, m-1])$

$IF (A[i] == 1)$

$POW_1 = 1$

$O-C = T_REDUCE(POW_0)$

$1-C = T_REDUCE(POW_1)$

$IF (O-C == m-1 \text{ AND } 1-C == 1)$

$PRINT("DA")$

$ELSE$

$PRINT("NE")$

8. $\rightarrow$ 4 ЗЕЛЕНА 2А РЕДОН

ЭЛЕВ

PAR(i in (0, m-1))

pon[i] = 1

PAR(i in (0, m-4))

for (j in (0, 3))

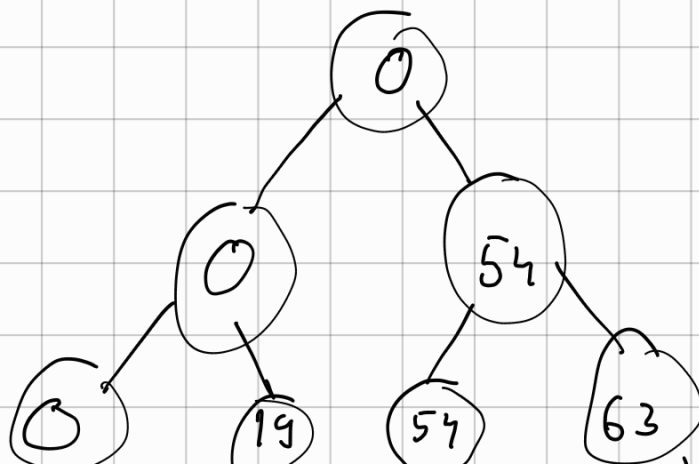
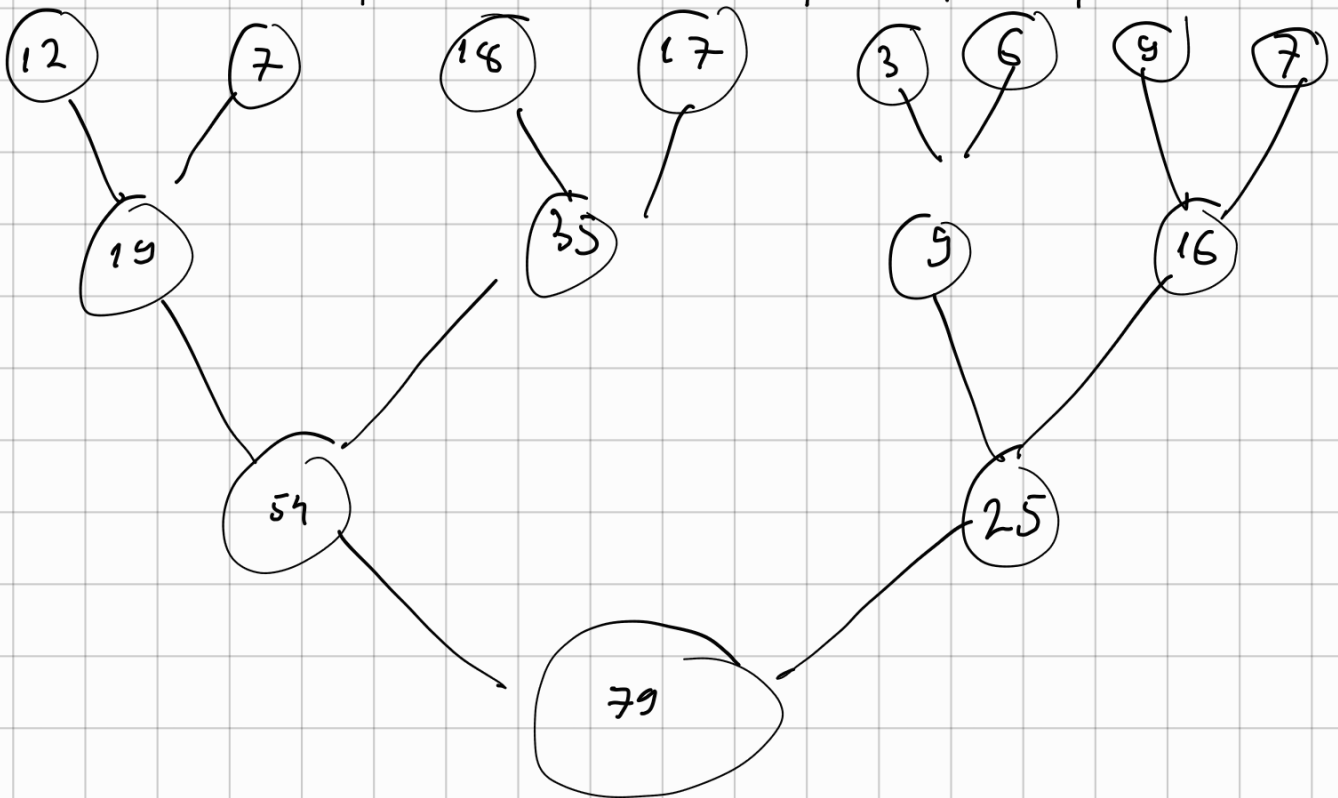
IF (P[i+j] != 1)

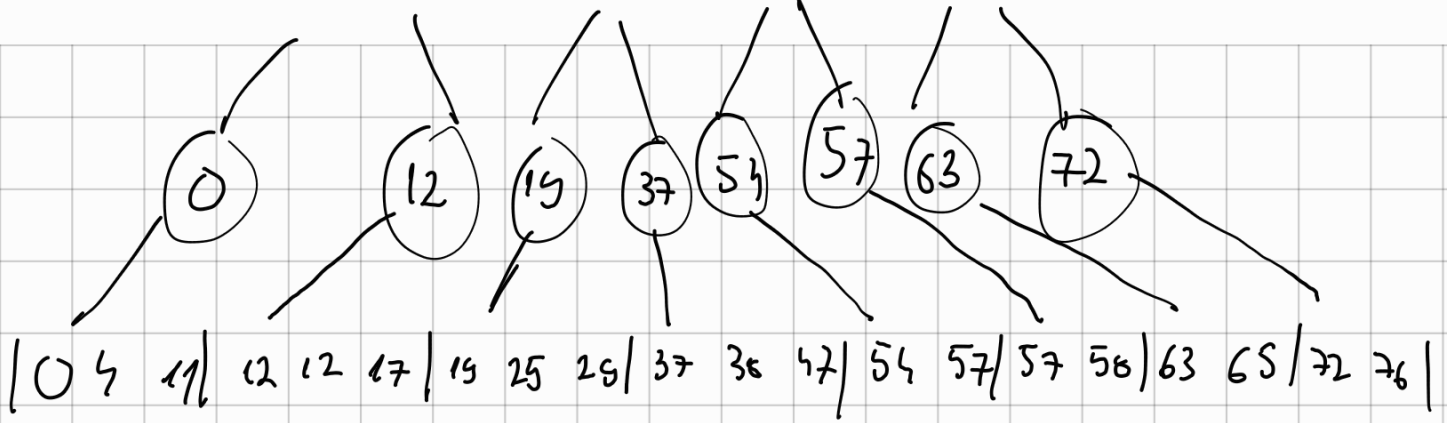
pon[i] = 0

RET = MAX-RECVCE (pon)

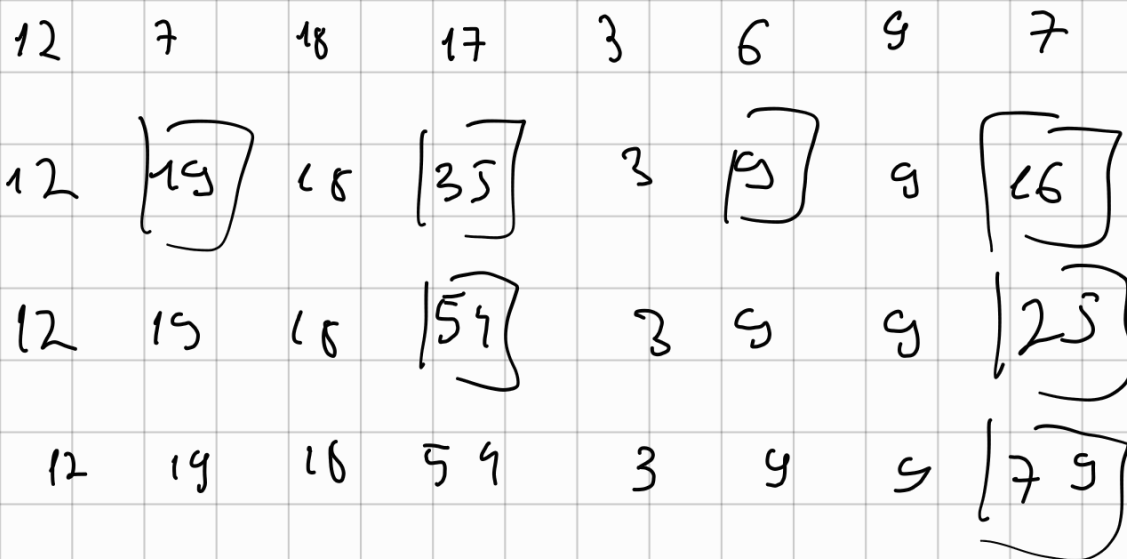
(4) 1. PRESCAN $p=8$ $m=20$ $20//8=2$ $20 \times 8 = 160 \Rightarrow$ PRIMA 4 SELE

1	2	3	4	5	6	7	8
4	7	1	0	5	2	6	4
8	1	9	7	3	0	1	5
2	7	7	3				

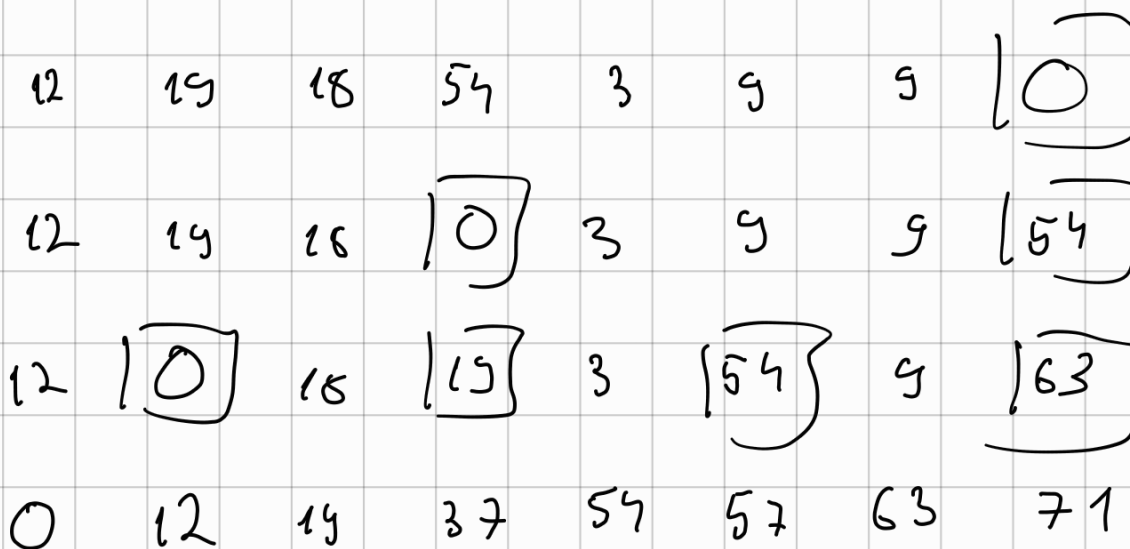




UPWARD



DOWN SWEEP



10.

PREW PLAN -)

PULSINA

(VA) DUWEG

MEPREKINUTOG

PODWIZIA

ISIRI

ZNAJENKI

PAR i in $(0, m-1)$

$POW[i] = 1$

$B[i] = P[i]$

$DODAJ[i] = 1$

PAR i in $(0, m-1)$

FOR j in $(1, m-1)$

$POW2[i] = POW[i]$

IF $(i+j < m)$

IF $(P[i] == B[i+j] \text{ AND } DODAJ[i] = 1)$

$POW[i] = POW2[i] + 1$

ELSE

$DODAJ[i] = 0$

$REF = MAX_REDUCE(POW)$

$O(m)$ -) NIJE DOBRO JER TAK IMA 1
SLIJEDEMI PROGRAM